



Mechanical and tribological properties of polyamide 6–polyurethane block copolymer reinforced with short glass fibers

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ABSTRACT

Reinforced polyamide 6–polyurethane (PA6–PU) block copolymers with different short glass fiber proportions were prepared by monomer casting (MC) technology. The mechanical and tribological behaviors of the reinforced composites were studied. The friction and wear experiments were tested under ambient conditions in a friction machine with the chromium steel ball sliding on the surface of the composites at rather high sliding speeds of 500, 1000 and 1500 r/min and loads of 78.5 and 157 N, respectively. Tensile strength of the composites increased with the addition of glass fibers, but notch impact strength decreased. The glass fiber reinforcement was found effective in reducing the coefficient of friction and the wear rate. The effect of the applied load and sliding speed on tribological properties of the composites was explored in this study. The load and sliding speed showed stronger effect on the wear rate of the composites. The results showed that both the coefficient of friction and wear rate values increased with the increasing of load. With the increasing of sliding speed, the coefficient of friction of the composites decreased, while the wear rate values increased. Microstructure of worn surface of the tested composites was inspected by scanning electronic microscope (SEM) and wear mechanism of the reinforced composites was studied.

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1. Introduction

Polyamide 6 (PA6) synthesized by anionic polymerization of caprolactam has been extensively investigated because of its high yield, high reaction rate, high molecular weight and crystallinity and excellent mechanical properties. However, the PA6 is notch sensitive, especially notch impact strength in low temperature. They are not resistant to crack propagation, and which often results in brittle or catastrophic failure [1,2]. Block copolymerization method [3,4] to improve toughness of PA6 has been applied widely, because the structure of block copolymer is stable and toughness of those is evident. For example, Sakurai et al. found that the soft segment has a low glass transition temperature, which gives the elastomeric character of polyamide 6–poly(propylene glycol) block copolymers at room temperature [5]. Polyamide 6–polyethylene glycol–polyamide 6 triblock copolymers synthesized by KAY JIN KIM showed the highest impact strength [6]. Many research workers' studies showed polyurethane soft segments embedded in molecular chains of polyamide block copolymer can improve the impact property of PA6 [7,8], such

as poly(ether urethane)–polyamide 6 synthesized by Seo [9,10]. In this paper, polyamide 6–polyurethane (PA6–PU) block copolymer was proposed and prepared to improve the impact strength of PA6, especially the impact strength at low temperature. However, the toughness and tribological properties of the copolymer were required at the same time in railway industrial fields. Many studies focused on the thermal stability, structural morphology and mechanical properties of the copolymers, but not on their friction and wear properties [11–13]. Furthermore, the effect of the reinforcing agents such as carbon [14] and glass fibers [15,16] on the friction and wear of the copolymers has been given little attention. Therefore, For the purpose of improving properties of the copolymer further, especially the friction and wear properties of the copolymer, it was decided to reinforce the copolymers with short glass fibers in this study. The mechanical and tribological properties of the copolymer and the copolymers with short glass fibers will be reported in this work.

PA6 and the PA6–PU block copolymer with 10% polyurethane (PU) soft component were synthesized by monomer cast (MC) technology, on the basis of which, different contents of short glass fiber reinforced copolymers were synthesized by the same technology. Its compatibility between glass fiber and copolymer was increased by the reaction between isocyanate group of PU soft phase and hydroxyl group on the surface of the glass fiber. The objective of this study is to study the mechanical and tribological properties of the

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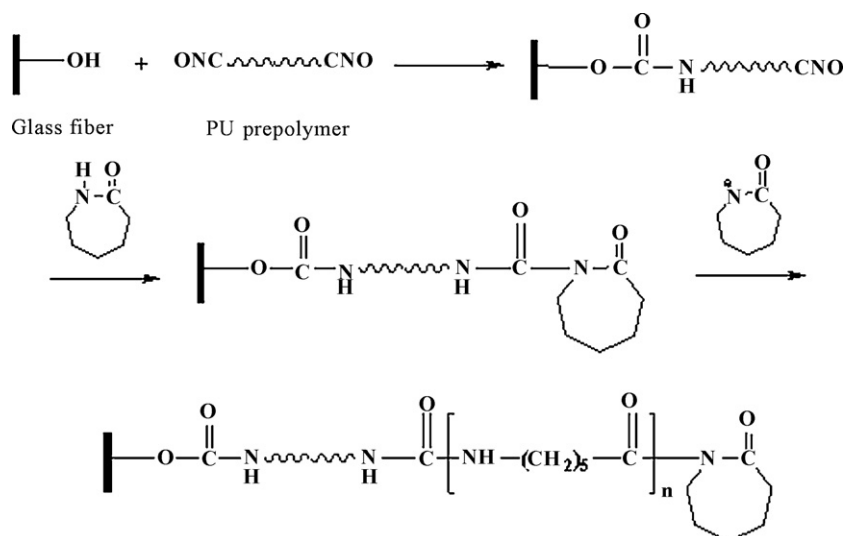


Fig. 1. The schematic diagram of GF surface reaction.

short glass fiber reinforced copolymers and to evaluate the influence of sliding speed and applied pressure values on the friction and wear characteristics of the composites.

2. Experimental details

2.1. Reagents

Caprolactam (CL) was purchased from Baling Co. China Petroleum and Chemical. It was dried under a vacuum at 40 °C for 12 h before use. Polyether glycerol (PPG-330), average molecular weight $M_n = 3000$, was supplied by Zhuzhou Times New Material Technology Co. Ltd. It was vacuum dried at 100 °C for 2 h prior to use. Diphenyl methane-4,4' diisocyanate (MDI) was supplied by Zhuzhou Times New Material Technology Co. Ltd. It was stored in ice-box before use. Sodium caprolactamate was supplied by Hunan HUIHONG Co. Ltd.

Short glass fiber was E-glass roving (fiber diameter is 10 μm and slenderness ratio is 10:1) was supplied by Nanjing Institute of Glass fiber Research and Design, China.

2.2. Preparation of materials

PU prepolymer was prepared at 80 °C in a water bath. A 500 ml flask equipped with a magnetic stirrer and a vacuum

pump was charged with n mol of MDI. After the complete melting of MDI, $2n$ mol of PPG was added with vigorous stirring. The reaction lasted for 1 h. Then the PU prepolymer with isocyanate group was obtained. It was kept in the ice-box before use.

According to the MC technology, there are two reaction vessels (A and B) with industrial pump and heating system. Sodium caprolactamate and CL were melted in vessel A at 130–140 °C. The PU prepolymer, untreated glass fiber (1%, 2%, 3% and 5%) and CL were added in vessel B at 130–140 °C. Fig. 1 shows the schematic diagram of glass fiber surface reaction. Once the prepolymer and catalyst were fully melted, both solutions in vessel A and B were mixed and poured into a preheated mold at 170 °C for 30 min, where the polymerization occurred. The mold was cooled down to 25 °C and the glass fiber reinforced PA6–PU block copolymer was released from the mold.

2.3. Mechanical testing

Izod notched impact test was measured according to Chinese Standard-T1043-1993, using an IZOD machine Model XJJ-50 using the pendulum method at 23 °C. The tensile tests were performed on a WDW electron universal material testing machine.

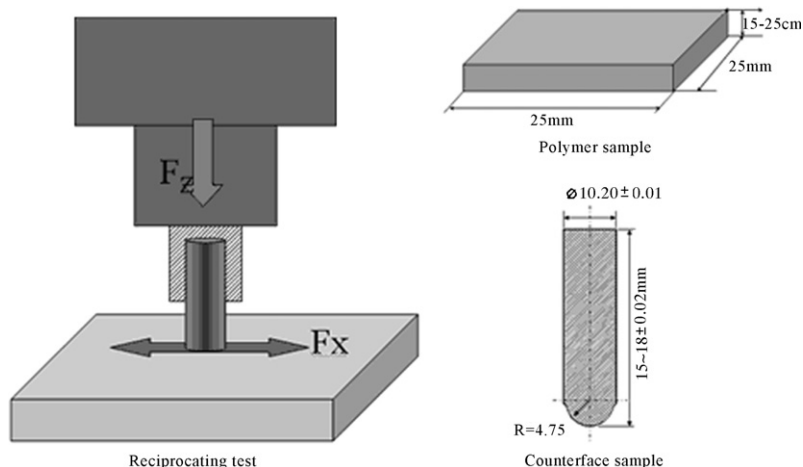


Fig. 2. Schematic diagram and sample size for the frictional couple. F_z – load direction; F_x – friction direction.

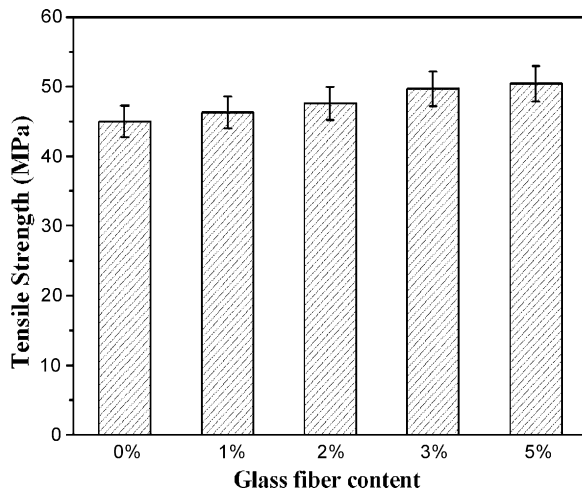


Fig. 3. Tensile strength of GF reinforced PA6-PU block copolymers in the proportions stated.

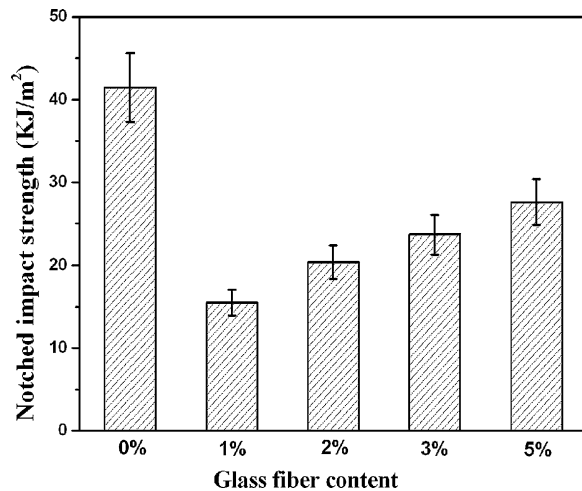


Fig. 4. Notched izod impact strength of GF reinforced PA6-PU block copolymers in the proportions stated.

2.4. Friction and wear analysis

The friction and wear tests were conducted on an UMT-3 (Universal Macro Materials Tester) friction and wear tester. An

assembly diagram of the frictional pairs is shown in Fig. 2. Columnar structure chromium steel with a $\Phi 9.5$ ball (HRC 62) was used as the counterface. The polymer specimen blocks were of 25 mm $\times$ 25 mm $\times$ 15 mm. Before each test, the blocks and the balls were ultrasonically cleaned in acetone and thoroughly dried. Reciprocating slide was performed under ambient conditions (temperature: $20 \pm 5^\circ\text{C}$, humidity: $50 \pm 10\%$). The test was performed with three speeds: 500, 1000 and 1500 r/min (one motion period of forward and backward is defined as one round, and the trace of friction is 10 mm) and with two loads: 78.5 and 157 N. The test durations ranged from 0 to 60 min. The average values of friction coefficient in the range from 40 to 60 min were used as the friction coefficient of samples.

Weight loss measurements were made using an analytical balance having an accuracy count of 10^{-4} g. Specific wear rate was calculated using the following equation:

$$\text{wear rate} = \frac{\Delta m}{\rho L d} \times 10^3 \text{ mm}^3/\text{Nm}$$

where Δm is the mass loss in grams, ρ the density of the test material in g/cm^3 , L the load in Newton and d is the sliding distance in meters.

2.5. Analysis of worn surfaces

Microstructures of the worn surface of samples sputtered with gold were inspected using a KYKY-2800 scanning electronic microscope (SEM). The aim is to study the wear track and to analyze the abrasion mechanism.

3. Results and discussion

3.1. Mechanical properties

The effect of glass fiber reinforcement on the mechanical properties of the PA6-PU block copolymer was studied. Fig. 3 shows the tensile strength of the glass fiber reinforced composites. With the increasing of glass fiber content, the tensile strength of the composites increased, however, the improvement of glass fiber on the tensile strength of the composites was not obvious according to the scope of experimental error. When the content of glass fiber increased to 5%, the tensile strength increased from 45 to 50.41 MPa, 12% increased. The possible reason was that the adhesion properties between glass fiber and polymer matrix in the composite was excellent and the dispersion degree of glass fiber in the composite was high.

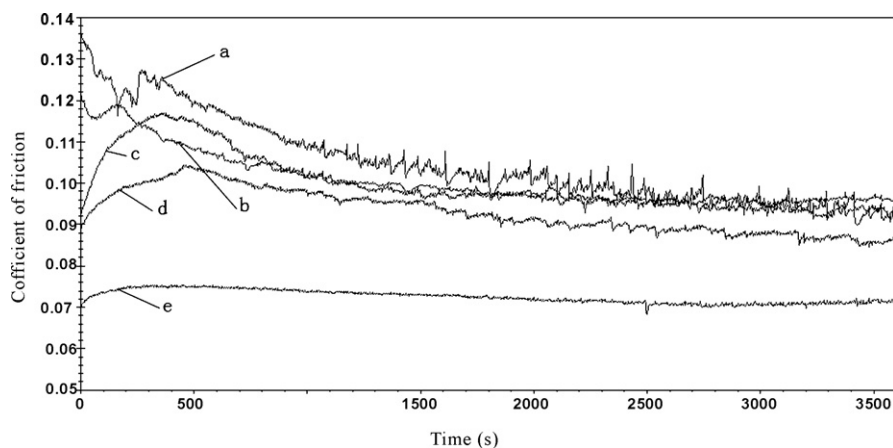


Fig. 5. Effect of sliding time on coefficient of friction of glass fiber reinforced PA6-PU block copolymers at a normal load of 78.5 N and sliding speed of 1000 r/min. (a) – 0%, (b) – 1%, (c) – 2%, (d) – 3%, (e) – 5%.

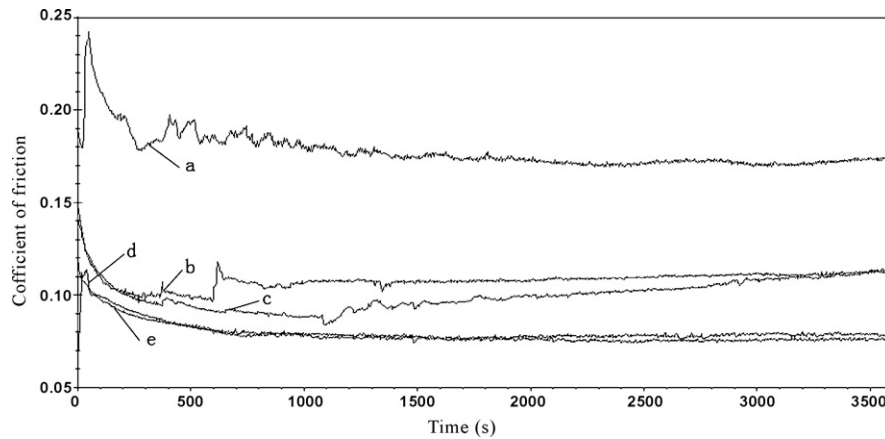


Fig. 6. Effect of sliding time on coefficient of friction of glass fiber reinforced PA6-PU block copolymers at a normal load of 157 N and sliding speed of 1000 r/min. (a) – 0%, (b) – 1%, (c) – 2%, (d) – 3% and (e) – 5%.

The Izod notched impact strength of the glass fiber reinforced PA6-PU block copolymer is shown in Fig. 4. When there was no glass fiber reinforcement in the copolymer, the impact toughness was excellent. When this copolymer was impacted, the PU soft component became the stress concentration point firstly, and then diffused into matrix, and the impact energy was consumed. It is the reason for the high impact strength of the PA6-PU block copolymer. However, when the glass fiber was added to the PA6-PU block copolymer, the impact strength of the composites decreased evidently. When the additive weight percent was 1%, the impact strength decreased from 41.48 to 15.51 kJ/m² which was a 62.6% decrease. However, with increasing of glass fiber content, from 1 to 5%, the impact strength increased a little, from 15.51 to 27.6 kJ/m². It is probably due to the present of glass fiber packed by PU soft phase structure which were caused by the reaction between hydroxyl group on the surface of glass fiber and isocyanate group of PU soft phase, which can improve the toughness of the composites.

3.2. Tribological properties

Fig. 5 shows the variation of coefficient of friction with sliding time at a normal load of 78.5 N and sliding speed of 1000 r/min for PA6-PU block copolymers reinforced with different proportion of glass fiber. The composites exhibited a lower coefficient of friction compared to PA6-PU block copolymer with the increasing of glass fiber reinforcement. At preliminary stage of friction, coefficient of friction of the composites increased a little. The reason for this phenomenon was that the surface of metal ball was not very smooth and the ploughed function on friction test caused by the bulge on surface of the ball was large. At the beginning, the transfer film was not formed between the metal counterpart and the composites. However, as the time of test went by, the stable transfer film was formed; coefficient of friction declined and tended to stable stage. It showed the same phenomenon in Fig. 6 about the variation of coefficient of friction of the composites with sliding time at

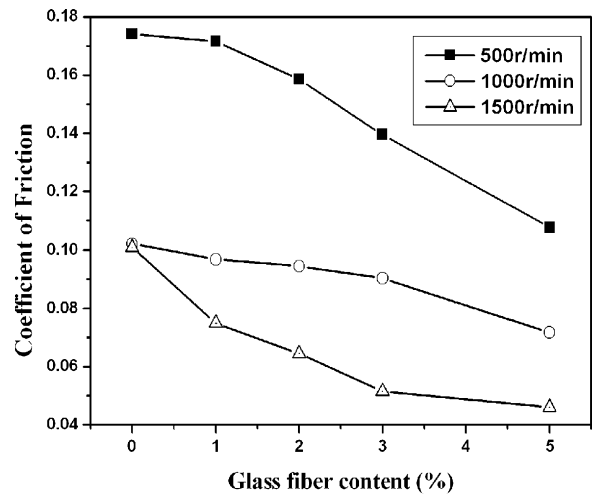


Fig. 7. Coefficient of friction of the composites under different sliding speed, 500, 1000 and 1500 r/min.

a normal load of 157 N. However, the stable stage appeared earlier at the higher load due to the shorter time of running-in stage.

Table 1 shows the data of coefficient of friction and wear rate of the composites under both loads. In the process of friction test, the

Table 1

Coefficient of friction and wear rate of the GF reinforced PA6-PU block copolymers under different loads.

Samples	Coefficient of friction		Wear rate ($\times 10^{-5}$ mm ³ /Nm)	
	78.5 N	157 N	78.5 N	157 N
GF-0	0.102	0.171	3.54	17.7
GF-1	0.096	0.109	3.24	7.02
GF-2	0.094	0.108	1.48	7.54
GF-3	0.090	0.089	1.47	4.87
GF-5	0.071	0.077	1.10	4.08

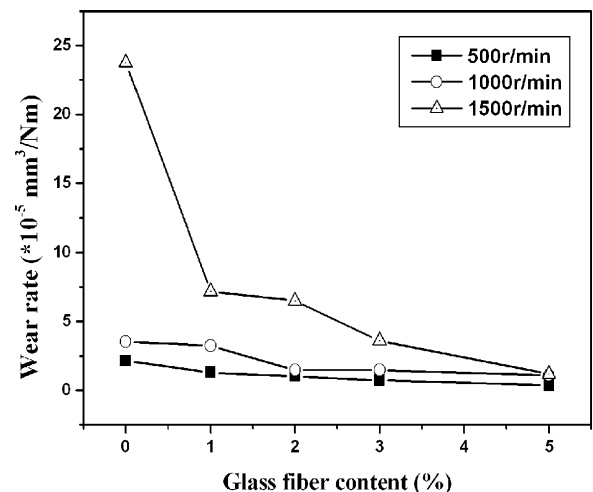


Fig. 8. Wear rate of the composites under different sliding speed, 500, 1000 and 1500 r/min.

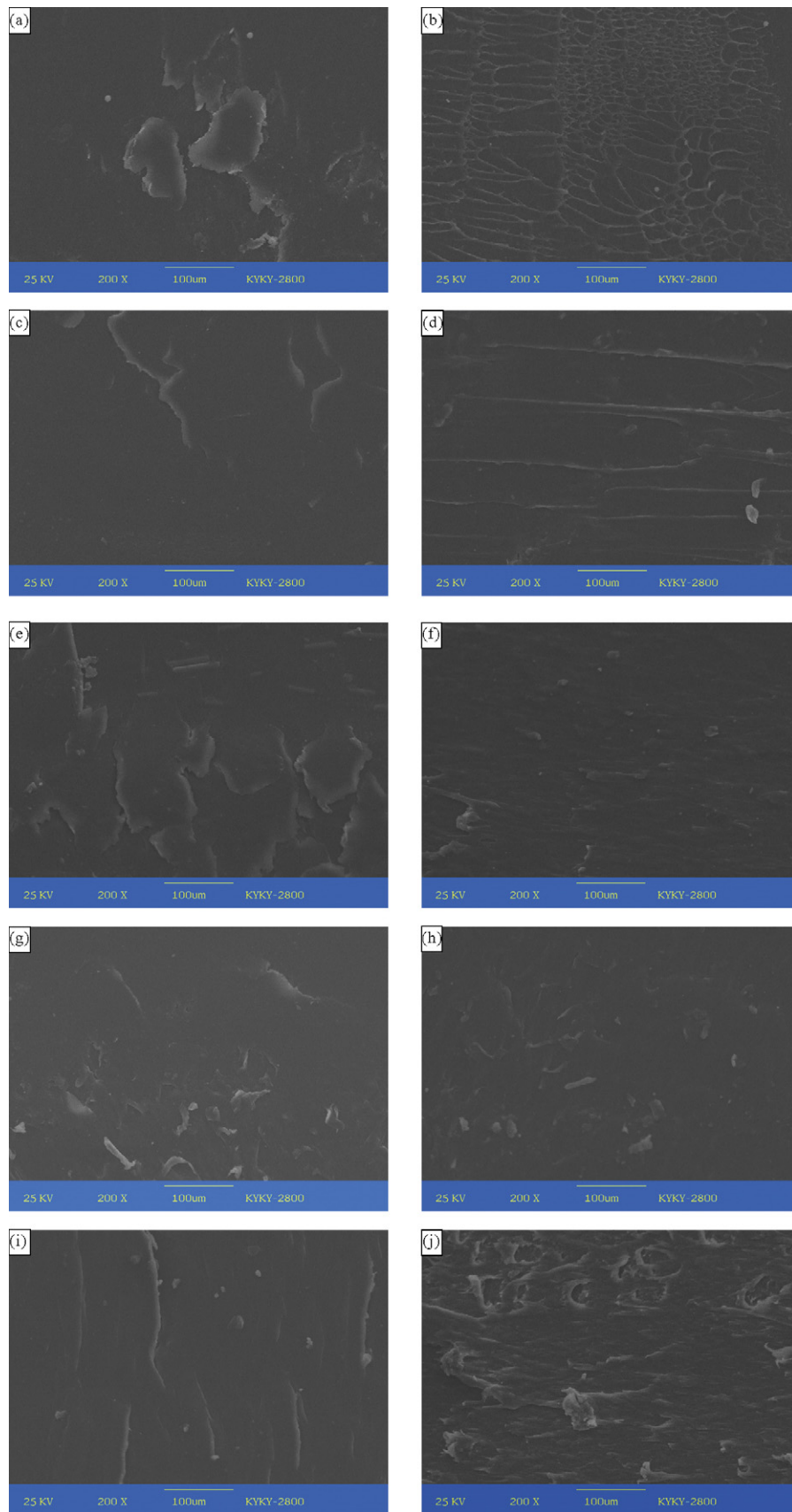


Fig. 9. SEM micrographs of worn surfaces of the composites under different loads, at a sliding speed of 1000 r/min: (a) 0%GF, 78.5 N, (b) 0%GF, 157 N, (c) 1%GF, 78.5 N, (d) 1%GF, 157 N, (e) 2%GF, 78.5 N, (f) 2%GF, 157 N, (g) 3%GF, 78.5 N, (h) 3%GF, 157 N, (i) 5%GF, 78.5 N and (j) 5%GF, 157 N.

matrix in the composites was dragged out by the metal ball. The glass fiber in the composites appeared gradually and shared partial pressure, and the actual area of contact decreased. So the coefficient of friction of the composite was reduced with the increasing of glass fiber content. It will be explained further in analysis of worn mechanism of the composite according to SEM photos.

Wear rate of the composites under different loads is shown in Table 1. Wear rate values of the composites decreased with increasing of glass fiber content at both loads. The mechanical properties of the composites were improved by glass fiber and the actual area of contact decreased in the process of friction. Adhesion and deformation of the composites were reduced. So the wear rate values decreased. Furthermore, heat conduction and heat stability of the composites were improved with increasing of glass fiber content, which made the resistance to abrasion better.

With the increasing of load, the coefficient of friction and wear rate of the composites increased. It can be illustrated by the increasing area of contact between the metal counterpart and the composites which induced higher flash temperature, heat generation, and the viscous-elastic property in the response of material stress, adhesion and transferring behaviors was influenced. Hence, the coefficient of friction and wear rate increased with the increasing of applied load. Another reason was that as the load increased, the glass fiber in the composites revealed earlier in the friction process, and it was not pulled out easily due to the excellent interfacial properties. So the coefficient of friction increased. The glass fiber on surface of the composite was crushed easily under the higher load, and stripped gradually. The wear rate increased dramatically.

Coefficient of friction and wear rate of the composites at different sliding speed are shown in Figs. 7 and 8, respectively. With increasing of sliding speed, coefficient of friction of the composites decreased. It can be explained by the change of temperature on the surface of the composites. The higher of the sliding speed of test, the faster of the accumulation of frictional heat in the friction interface, thus leading to partial surface melting, the lubricate component provided by melting polymer matrix lead to the decrease of the coefficient of friction. For glass fiber reinforced composites, it was due to the rubbing of glass fiber of faced materials which caused a drop in coefficient of friction.

It is clear from Fig. 8 that the wear rate values almost increased with the increasing of sliding speed for all composites. The reason for this behavior was that the friction caused surface temperature to increase. High wear rates were caused by increase in wear rate associated with the surface layer reaching the softening point of the composites. As it is known that wear process in polymers involve adhesive and plastic flow, transitions between regions dominated by each of these commonly lead to changes in wear rate with applied pressure and sliding speed. Furthermore, this result was closely related to structure characteristics, and chemical effects occurred in frictional processes as well as transfer film formation on the counterpart material. At the same time, the adhesive properties between the glass fiber and the matrix in the composite became unimportant due to melting polymer matrix caused by higher flash temperature on the surface of the composite. The shedding of glass fiber leads to increase of the wear rate. It was said that the presence of glass fiber results in some abrasive mechanism besides the adhesive wear mechanism.

3.3. SEM observation of the worn surfaces

Fig. 9 shows SEM micrographs of the worn surfaces of the composites under 78.5 and 157 N loads. Under the condition of 78.5 N load, flakes appeared on the wear scar of no glass fiber reinforced PA6–PU block copolymer, which indicated that adhesive wear played an important role. It was related to the thermal and mechanical properties of the copolymer. When the glass fiber content was 1%,

shown in Fig. 9(c), the worn surface of the composite was smooth. It was due to the mechanical properties of the composite improved by addition of glass fiber. The friction performance of the composite was better than that of copolymer. The abrasive particle appeared gradually and some pits formed by abscission of the fibers were also found with increasing of glass fiber content, and the exposed glass fiber on the surface of the composite was found in Fig. 9(e) and (g). The glass fiber exposed on the surface of the composite played a role of sharing pressure. On the other hand, the cutting effect of the bulge on surface of the metal ball on the polymer matrix of the composites was impeded by the exposed glass fibers. When glass fiber content was 5%, shown in Fig. 9(i), the separation between glass fiber and matrix was caused by surface stress, and the crumb was generated. The silver-stripes on the surface of the composite were formed in the process of friction which suggested the matrix of the composites was shifted by reason of heat adhesive function. It showed that both abrasive wear and adhesive wear played important role in explaining of wear mechanism.

It can be seen that the wrinkled with wavy morphology appeared on the wear scar of the unreinforced block copolymer under 157 N load in Fig. 9(b). It was due to the heating effect of friction which promoted a temperature rise hence, softening the polymer surface, and as a result, the wear regime became a severe surface with wrinkling occurrence. With the addition of glass fiber, shown in Fig. 9(d), the plucked and ploughed marks, appeared on the surface of composites, indicated that adhesive wear was primary wear mechanism which can explain the friction and wear phenomenon. Fig. 9(f) and (h) shows that the abrasive dust and some pores formed by the extract or fracture of glass fiber were found with increasing of glass fiber. The hardness of the composites became larger with the increasing of the glass fiber content. The maximum compressive stress in the contact plane of the composites was more than the strength of damaging abrasive particles, and the particles were crushed constantly, then the crushing abrasive wear appeared. It suggested that the transition from adhesive wear to abrasive wear of wear mechanism occurred. When the glass fiber content was up to 5%, the glass fibers in the composite were pulled out heavily. The fatigue deformation on the surface of composites appeared, which indicated the fatigue wear played an important role, and accompany with adhesive and abrasive wear.

4. Conclusion

The short glass fiber modified PA6–PU block copolymers were proposed and prepared by means of MC technology firstly. The compatibility between glass fiber and the copolymer was improved by the chemical reaction of hydroxyl group on the surface of glass fiber and PU prepolymer with isocyanate group. The tribological properties of the composites were studied mainly and the conclusion of test was as follows:

- 1 When the copolymer was reinforced with glass fiber, the tensile strength of the composites increased with the increasing of glass fiber content. However, the notch impact strength first decreased. Then, the impact strength increased a little with the increasing of the glass fiber reinforcement.
- 2 With the addition of glass fiber, coefficient of friction and wear rate of the composites decreased. For example, under low load, when the glass fiber content was 5%, the coefficient of friction decreased 30%, and the wear rate decreased 68%.
- 3 The influence on coefficient of friction and wear rate by load and sliding speed was studied in this work. Coefficient of friction and wear rate values of the composites increased with the increasing of applied load. With increasing of sliding speed, coefficient of friction of the composites decreased and wear rate values

increased. The load and sliding speed showed stronger effect on the wear rate of the composites.

- 4 Worn surfaces of the tested materials showed that the adhesive wear played an important role for the block copolymer. With the addition of glass fiber, the wear mechanism was a combination of adhesive and abrasive wear. At the condition of high load, fatigue wear played an important role in the friction process.

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